

Selina

AWS Compute Grant Proposal, OpenCV AI Competition 2026, powered by AWS

Team name

Moses

Path pursued

Agentic Vision Award. Cloud-Optimized Vision with COOL is a stretch goal, added later if build time allows.

Problem statement and real-world impact

Selina is a multi-agent system built to support women through safety, health, work, education, and everyday life. It is dedicated to two women lost during pregnancy and childbirth, and it exists to give women more support, better information and someone to turn to when they need help.

One part of Selina, the Safety Agent, helps a woman plan a journey or meeting, checks in on her, and organizes an incident timeline if something goes wrong. Right now that timeline is built from what she reports in words. In practice, women often also have a photo or a short video from the moment something felt wrong, and that material tends to sit unorganized on a phone, hard to use quickly, and sometimes containing other people's faces that should not be shared without care.

This proposal adds a real computer vision layer to the Safety Agent using OpenCV 5. When a woman submits a photo or clip, the system checks whether it is usable, blurs faces that are not hers before anything is shared further, flags unusual movement patterns worth a closer look, and pulls out the clearest frame for the record. What the vision step finds changes what Selina does next, it can move an item higher in priority, ask for a retake, or prepare a clearer escalation for human review. The impact is a woman spending less time organizing evidence in a stressful moment, and a clearer, more protected record if she chooses to escalate.

Planned OpenCV 5 image or video analysis

- Face detection and automatic blurring of bystanders and other third parties before any image leaves the pipeline, so the woman's evidence does not expose people who did not consent to being recorded.
- Image and frame quality checks (blur, exposure, framing) that flag unusable submissions and prompt a retake instead of silently keeping a bad photo in the record.
- Motion pattern analysis on short video clips using frame differencing and optical flow, to flag sudden or repeated movement worth prioritizing for human review. This is a prioritization signal, not a claim of what happened.
- Key frame extraction from video, so the incident timeline stores one clear, representative image rather than a raw clip nobody has time to review.

- Basic scene context detection (indoor or outdoor, presence of a vehicle) to help tag the record without identifying the woman herself or monitoring anyone covertly.

All of this runs on the actual OpenCV 5 pipeline, not a wrapper around a third-party vision API, and the output is what determines the Safety Agent's next recommended action.

Planned AWS architecture and services

- Amazon S3 for encrypted, time-limited storage of submitted photos and video, deleted automatically after a short retention window unless the woman chooses to keep it in her case timeline.
- OpenCV 5 processing deployed on AWS Graviton (Arm), either an EC2 Graviton instance or a Lambda container image, so the core vision workload runs on the Arm path.
- API Gateway plus Lambda as the entry point that receives a submission and triggers the vision pipeline.
- IAM roles scoped tightly per function, and encryption at rest through AWS KMS for anything containing personal images.
- CloudWatch for basic latency and error observability during evaluation.

High-level architecture

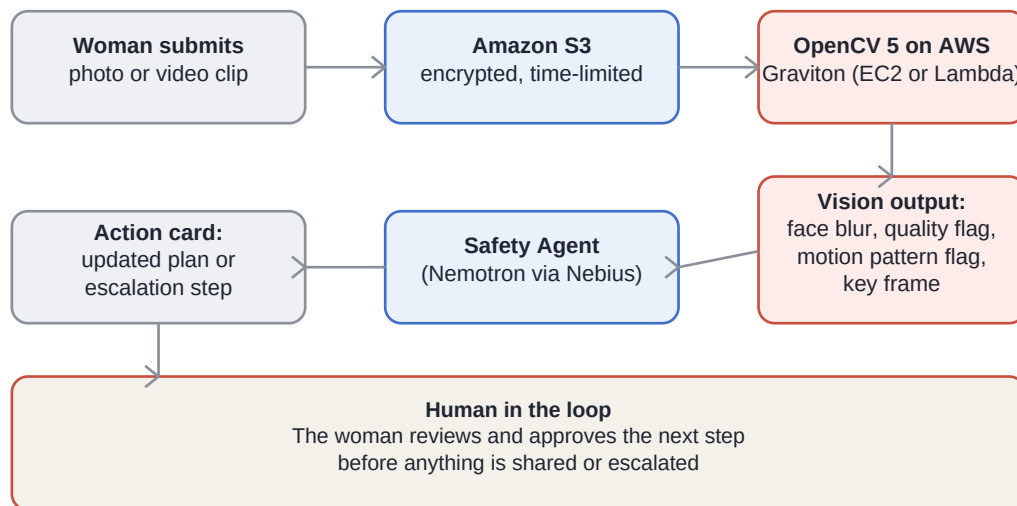


Figure 1. A submitted photo or video moves through OpenCV 5 on AWS Graviton, into the Safety Agent's reasoning, and ends with the woman approving what happens next.

Target users and beneficiaries

The same women Selina is built for: students, domestic workers, graduates and professionals who face safety risk in daily life, including on journeys, at work, or in unfamiliar situations, and who need a faster, calmer way to organize evidence and get to the right kind of help.

Proposed evaluation method and judge demonstration

- A set of synthetic and consented demo photos and clips, never real incident material, used to show face blurring, quality flagging and motion pattern detection before and after.
- Reproducible latency and throughput measurements for the Graviton pipeline, reported against a baseline run on a standard x86 instance.
- A live or recorded walkthrough where a submitted clip moves through the pipeline in front of judges and visibly changes the Safety Agent's next action, showing the perception to decision to action loop rather than a static result.
- A short review of failure cases, such as a low quality submission correctly triggering a retake request instead of a false pattern flag.

Team bio

Moses, software engineer, AI developer, and web3 builder. Building solo for now, full time, with a team possibly joining before submission.